Supplier Capability and Tooling Cost Study

Pressed and fabricated brackets - process selection against annual volume

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Volume basis	FY25 forecast, 18,000 units platform volume
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Summary

Many value-engineering proposals assume a process route without testing whether it is the right one at the relevant volume. This study sets out the cost structure of fabrication, progressive pressing and forging for chassis brackets across the qualified supplier panel, and identifies the volume at which each becomes preferable. For the front spring bracket the break-even between fabrication and a progressive press tool falls at 21,400 units per year, above the current platform volume. Several proposals closed on cost grounds would become viable if assessed at combined platform volume rather than single-programme volume.

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1 Scope and supplier panel

This study covers structural brackets in the mass range 1.5 kg to 8 kg, which accounts for 84 per cent of the bracket population on the HCV platforms. It excludes castings, which are covered separately, and excludes brackets integrated into larger weldments.

1.1 Panel composition

Capability	Qualified sources	Max part mass	Lead time, new tool
Fabrication and welding	7	No practical limit	6 to 8 weeks
Progressive pressing	4	6.0 kg	22 to 30 weeks
Transfer pressing	2	11.0 kg	26 to 34 weeks
Closed-die forging	3	No practical limit	30 to 40 weeks
Robotic welding cell	3	-	10 to 14 weeks

1.2 Capability constraints that affect design

- No panel source can progressive-press above 6.0 kg finished mass. Proposals that move a heavier part to pressing require a transfer press source.
- Only three sources operate a robotic welding cell. A fabrication proposal that depends on robotic weld cost must name a source that has one.
- Tool lead time for pressing and forging exceeds the normal running-change window. Proposals requiring new hard tooling are platform changes by definition, whatever their technical content.

Finding 1.1. Thirty-three ideas were closed in FY24 H1 for lack of supplier capability. In 19 of those the capability existed on the panel but not at the source currently making the part. Proposals should state the capability required rather than assuming the incumbent source.

2 Cost structure by process

Cost element	Fabrication	Progressive press	Forging
Hard tooling, typical	INR 180,000	INR 3,850,000	INR 7,200,000
Tool life, strokes	Not applicable	1,200,000	40,000 per die set
Material utilisation	68 to 74%	76 to 82%	88 to 94%
Cycle time	6 to 11 min	4 to 7 s	22 to 30 s
Labour content	High	Very low	Low
Scrap rate, steady state	2.4%	0.8%	1.6%
Typical piece cost, 4 kg part	INR 486	INR 318	INR 266
Secondary operations	Dress, straighten	Pierce, form	Trim, heat treat

Fabrication carries almost no tooling and a high variable cost. Forging carries high tooling and the lowest variable cost. Pressing sits between the two. The consequence is that the correct process is a function of volume and of nothing else, once the part is within the capability envelope of all three.

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3 Volume sensitivity and break-even

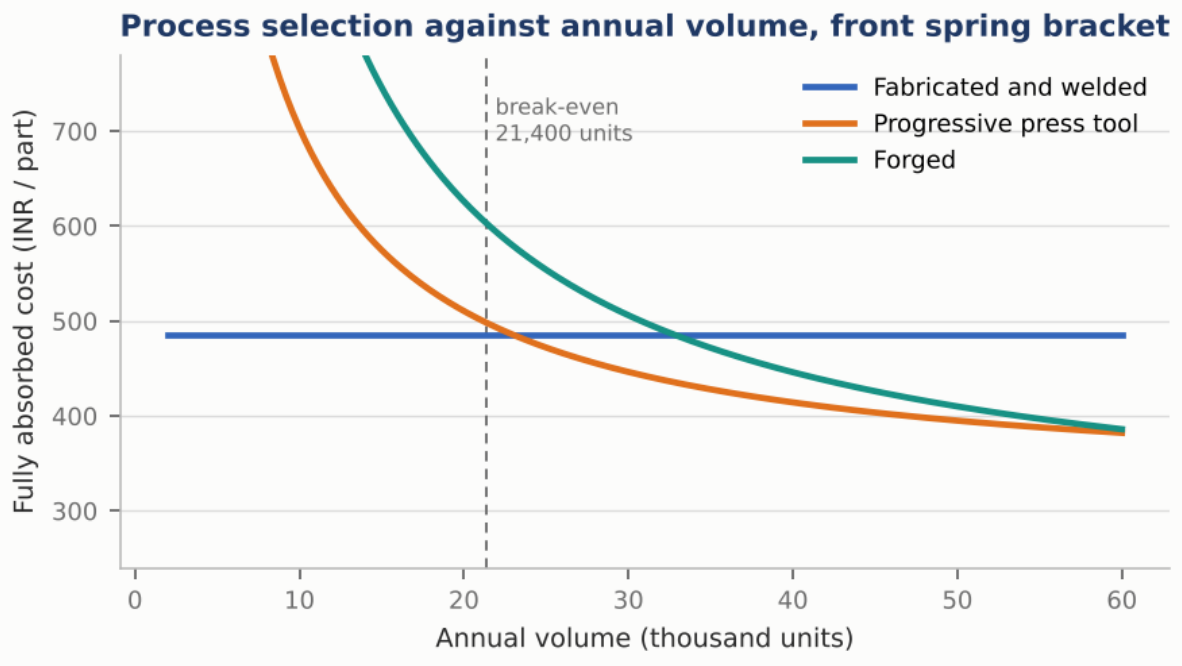


Figure 1 - Fully absorbed cost per part against annual volume for the three process routes. Tooling is amortised over four years. The break-even between fabrication and progressive pressing falls at 21,400 units per year.

Annual volume	Fabrication	Progressive press	Forging	Lowest cost
5,000	INR 486	INR 1,088	INR 1,706	Fabrication
10,000	INR 486	INR 703	INR 986	Fabrication
18,000 (current)	INR 486	INR 532	INR 666	Fabrication
21,400 (break-even)	INR 486	INR 486	INR 603	Equal
30,000	INR 486	INR 446	INR 506	Pressing
45,000	INR 486	INR 404	INR 426	Pressing
60,000	INR 486	INR 382	INR 386	Pressing

Finding 3.1. At the current platform volume of 18,000 units, fabrication remains the lowest-cost route for this part class. The margin is INR 46 per part, which is within the estimating tolerance. Proposals to move this class to pressing are neither clearly right nor clearly wrong at current volume and should be decided on lead time and quality rather than on cost.

3.1 Combined platform volume

The break-even assumes single-programme volume. Where a bracket is common across the rigid, tipper and tractor-trailer programmes, the combined volume is 29,400 units, which is above the break-even. Several proposals closed on cost grounds were assessed at single-programme volume and would reverse at combined volume.

Finding 3.2. Commonisation and process change interact. A commonisation proposal that is marginal on its own may carry a process change that is clearly beneficial at the combined volume. Assessing the two separately understates both.

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4 Worked example: front spring bracket

The front spring bracket is used to illustrate the full assessment because it appears in several open proposals and because its teardown comparison is documented in CVOEM/VAVE/TD-BM/2024/015 section 6.

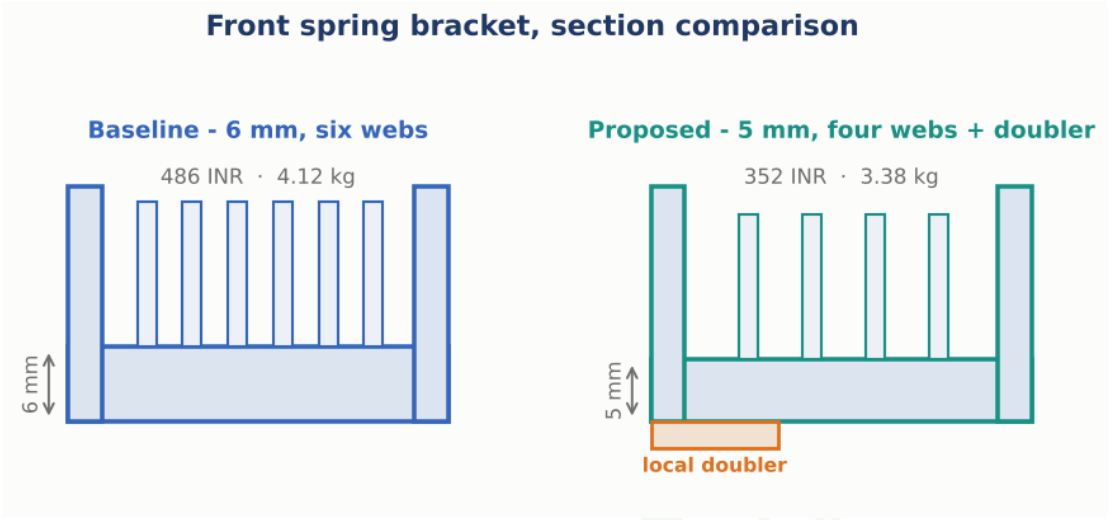


Figure 2 - Baseline fabricated bracket against the proposed thinner section with a local doubler. The doubler recovers section modulus at the highest-stress corner.

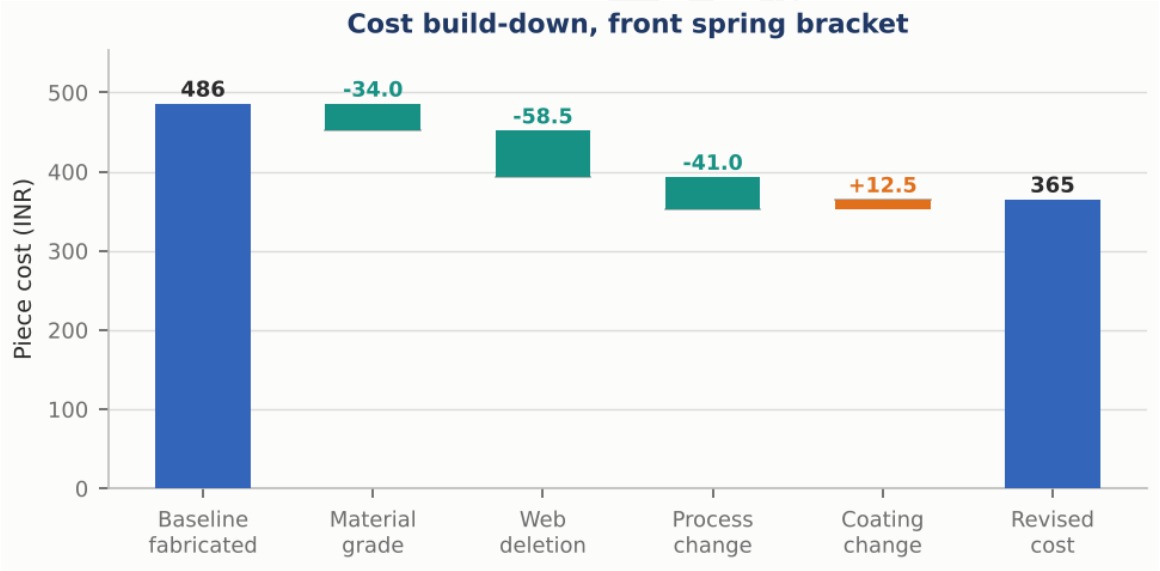


Figure 3 - Cost build-down for the front spring bracket. The coating change adds INR 12.5 and is required to meet the exposure class C1 requirement in CVOEM/VAVE/ES-STD/2023/004 section 3.

Step	Change	Cost effect	Mass effect	Note
1	Baseline fabricated bracket	INR 486	4.12 kg	Current
2	HSLA 340 in place of IS 2062	-INR 34.0	0.00 kg	Enables step 3
3	5 mm section, four webs, doubler	-INR 58.5	-0.74 kg	Requires step 2
4	Robotic weld in place of manual	-INR 41.0	0.00 kg	Needs a cell source

Step	Change	Cost effect	Mass effect	Note
5	Zinc-nickel in place of 12 um zinc	+INR 12.5	0.00 kg	Required by standard
	Revised	INR 365	3.38 kg	

Finding 4.1. Steps 2 and 3 are not separable. Submitted individually, the HSLA upgrade shows a cost increase and the section reduction fails validation. Submitted together they deliver INR 92.5 and 0.74 kg. The idea register records them as two separate closed ideas.

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5 Implications for idea assessment

- A proposal that changes the process route shall state the volume at which it is assessed, and whether that is single-programme or combined.
- A proposal requiring new hard tooling is a platform change regardless of its technical simplicity, because tool lead time exceeds the running-change window.
- Where a proposal depends on a capability the incumbent source does not have, the proposal should name the required capability rather than being closed as not feasible.
- Dependent proposals should be submitted together. A material upgrade that enables a section reduction has no independent business case and will be closed if submitted alone.
- Cost estimates in proposals should use the process adders in CVOEM/VAVE/MC-IX/2025/003 section 3.2 so that alternatives are ranked consistently.

Recommendation. The committee is asked to adopt a combined-volume basis for process-change assessment on parts common to three or more programmes, and to re-test the nineteen capability closures identified in finding 1.1 against the full supplier panel.

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